

Wireless DIGITAL THERMOMETER For MULTIPLE SENSORS

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My last article 'Remote Sensing Thermometer' published in June 2013 issue gave the constructional details of a project for sensing and measuring temperature of a body from a remote place. The circuit described there works absolutely fine, but it suffers from a major drawback; it cannot measure more than one temperature at a time. To overcome this problem, a different technique is adopted here, which allows you to sense eight (at the most) transducers at a time.

The technique

Block diagram of the wireless addressable digital thermometer is shown in Fig. 1. At the transmitting end, an 8-bit analogue-to-digital con-

verter (ADC) continuously scans and converts signals from eight different temperature sensors. The sensors are selected sequentially by a 3-bit binary addressing system. At any moment, output of the ADC is an 8-bit binary number representing the analogue signal developed at the output of a particular temperature sensor, being addressed at that moment.

By using special parallel-to-serial encoders, this 8-bit data, along with the binary address of the sensor, is sent serially to the remote receiving end. Communication between the two ends are met with the help of a pair of 433MHz UHF transmitter and receiver modules operating in ASK/OOK mode.

At the receiving end, the transmitted signal is received by

a 433MHz ASK/OOK RF receiver module. The received 8-bit serial signal is then converted back to its original parallel form, by using special data decoders as explained later. An equivalent analogue signal is then developed from this data by an 8-bit digital-to-analogue converter (DAC). A digital multimeter connected at the output of the DAC is used to show the temperature on mV scale.

Sensors identification

Scanning of sensors is governed by a 3-bit binary counter, which places, in a cyclic fashion, sequentially-changing binary numbers from 000 to 111 on the three designated scanning input lines, A, B and C, of the ADC. Each number bears the binary address of a particular sensor, and whenever addressed, the sensor sends data to the ADC for conversion to its digital equivalent.

Here comes a problem. Due to continuous scanning of sensors by the ADC, received data may come from any one of the possible eight sensors. Hence, for proper identification of a source sensor, specific 8-bit binary number (address) is tagged with the 8-bit signal output of each sensor, prior to transmission. Special data encoders are used for this purpose.

The encoder used in this system can send only four bits of the 8-bit sensed data at a time. Hence, the sensed 8-bit signal from the ADC is first broken into two nibbles of four bits each. The nibbles are then sent, one after the other, repeatedly to the receiving end, after attaching the said 8-bit address to each of these by the said encoders.

Among the total eight bits of the address word, three are scanning bits, as explained above, and fixed for a

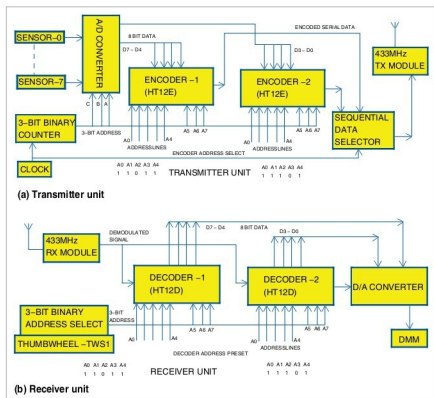


Fig. 1: Block diagram of the wireless digital thermometer